

Aakash Bhat

Software Engineer / DevOps & Automation

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EDUCATION

•J.C. Bose University of Science & Technology, YMCA, Faridabad

B.Tech. in Computer Engineering

2022 – 2026

•Delhi Public School, Faridabad

CBSE (Class XII)

2022

•Delhi Public School, Faridabad

CBSE (Class X)

2020

EXPERIENCE

•Earnest Tactical Solution Private Limited (ETSPL)

2026 – Present

Software Engineer, DevOps Engineer & Technical Consultant

India

- Work across software engineering, DevOps, and technical consulting for defence technology projects focused on technology indigenisation and field deployment.
- Develop AI-based detection and surveillance systems by integrating computer vision models, backend APIs, dashboards, and containerized services.
- Support technical evaluation, system integration, and solution documentation for UAVs, power electronics, and other defence applications.

•Codec Technologies Pvt. Ltd.

Sep 2025 – Dec 2025

Data Science and MERN Stack Developer Trainee

Remote

- Completed AICTE & ICAC approved training in data science and full-stack development, covering preprocessing, model development, REST APIs, and component-based interfaces.
- Built Python analytics workflows and MERN application components using MongoDB, Express.js, React, and Node.js.

•Genic Minds

May 2025 – Aug 2025

AI/Computer Vision Intern

Remote

- Developed facial recognition pipelines using deep learning, image embeddings, and OpenCV.
- Built livestock identity models and worked across dataset preparation, annotation, CNN training, and edge-device inference optimization.

PERSONAL PROJECTS

•Argus – Multi-Camera Intrusion Detection

2026

FastAPI, Next.js, OpenVINO, PostgreSQL, Docker, AWS EC2

GitHub

- Built a self-hosted surveillance platform that batches YOLO/OpenVINO inference across multiple camera streams and tracks people inside configurable regions of interest.
- Implemented contextual risk scoring, authenticated WebSocket alerts, WebRTC playback through MediaMTX, and tenant-aware APIs backed by PostgreSQL or SQLite.
- Containerized the FastAPI backend, Next.js dashboard, PostgreSQL, MediaMTX, and Nginx services with Docker Compose and deployed the stack on AWS EC2 behind an Nginx reverse proxy.

•GeMSentry – Government Tender Intelligence

2026

Python, Playwright, Flask, SQLite, PyPDF

GitHub

- Automated GeM tender discovery with configurable keyword searches, date filtering, PDF downloads, and extraction of EMD, exemption, pre-bid, and ePBG details.
- Built a Flask and SQLite review dashboard with bid scoring, status workflows, cached document access, authenticated APIs, pytest coverage, and Google Sheets synchronization.

•Panox V2 – Dual-Fisheye 360 Video Stitcher

Python, OpenCV, CUDA, PyTorch, GitHub Actions

2026

GitHub

- Engineered a dual-fisheye-to-equirectangular video pipeline with ORB/RANSAC alignment, adaptive seams, multi-band blending, batch export, and spherical metadata injection.
- Added hardware detection, optional CUDA/NVENC acceleration, automatic CPU fallback, configuration presets, and CI tests across Python 3.10 and 3.12.

•PhishGuard 2.0 Model Trainer

PyTorch, LightGBM, scikit-learn, GPU training

2026

GitHub

- Developed a phishing-URL training pipeline that harmonizes more than 1.2 million labeled records and extracts 30 lexical and domain-specific security features.
- Combined LightGBM, a PyTorch residual network, and character n-gram TF-IDF models through calibrated mixture-of-experts scoring with CPU fallback and FP16 GPU support.

•Multi-Modal Binary Malware Analysis

PyTorch, PyTorch Geometric, FAISS, Streamlit

2026

GitHub

- Implemented opcode Transformer, control-flow graph GIN, and system-call BiLSTM encoders with contrastive alignment and attention-weighted fusion.
- Added two-stage static and dynamic triage, missing-modality gating, FAISS similarity search, experiment-backed metrics, and a Streamlit research dashboard.

•PDFZen – Local Document Processing Suite

Vite, JavaScript, FastAPI, PyMuPDF, LibreOffice

2026

GitHub

- Created a privacy-focused PDF application with in-browser merge, split, compression, organization, rotation, cropping, watermarking, and page-numbering tools.
- Implemented a local FastAPI conversion backend for PDF-to-Word, Excel, PowerPoint, image, and Markdown workflows using pdf2docx, pandas, PyMuPDF, and LibreOffice.

TECHNICAL SKILLS AND INTERESTS

Languages: Python, JavaScript, TypeScript, SQL, HTML5, CSS3

Backend and Web: FastAPI, Flask, Next.js, React, Node.js, REST APIs, WebSockets

AI and Machine Learning: PyTorch, OpenVINO, OpenCV, YOLO, LightGBM, scikit-learn, FAISS, TensorFlow

Data and Infrastructure: PostgreSQL, SQLite, MongoDB, SQLAlchemy, Docker, Docker Compose, Nginx, MediaMTX, AWS (EC2)

Automation and Testing: Playwright, PyPDF, pytest, Jest, GitHub Actions, Postman

Core Concepts: Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, Machine Learning, Computer Vision

CERTIFICATIONS

•Regression Analysis: Simplify Complex Data Relationships

Google

Mar 2026

•The Power of Statistics

Google

Feb 2026

•Go Beyond the Numbers: Translate Data into Insights

Google

Feb 2026

•Foundations of Data Science

Google

Feb 2026

•Advanced R Programming for Data Analytics in Business

NPTEL (IIT Kanpur) | SWAYAM, Govt. of India

12-week Online Certification

Jul – Oct 2025